BUSINESS METHOD FOR DUAL-PURPOSE LOGISTICS NETWORKS

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Title: BUSINESS METHOD FOR DUAL-PURPOSE LOGISTICS NETWORKS

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Abstract

A business method for integrating and optimizing a primary logistics network (e.g., ride-hailing, package delivery) with a secondary data logistics network. The method comprises: receiving a primary transport request (e.g., passenger pickup, package delivery); identifying vehicles within the primary network capable of fulfilling the request; optimizing the routing of selected vehicles using network theory and directed graph searches to achieve the primary transport goal; and concurrently, during the optimized route, utilizing the selected vehicles to physically transport portable data cartridges to accomplish a secondary data logistics goal. This dual-purpose optimization reuses existing vehicle resources to create a highly efficient, integrated network for both physical items/passengers and high-volume data transfer.

Background of the Invention

Logistics networks, such as ride-hailing services, package delivery companies, and shipping lines, operate vast fleets of vehicles to move passengers and physical goods. These networks are optimized primarily for their main task: getting a person or package from point A to point

B efficiently. Simultaneously, there is a growing need for physical data transport services for massive datasets, as traditional telecommunication networks often become bottlenecks or pose security risks for such volumes. However, operating a dedicated data logistics network requires its own significant investment in vehicles and infrastructure. There is a need for a business method that can intelligently leverage existing logistics networks to efficiently perform data transport as a secondary function, thereby maximizing resource utilization and creating new revenue streams without requiring entirely separate infrastructure.

Summary of the Invention

The present invention is a business method for operating and optimizing dual-purpose logistics networks. The core idea is to integrate a high-volume physical data logistics service into an existing network that primarily transports passengers or physical goods.

The method involves a central optimization system that receives requests for primary logistics tasks (e.g., a customer requesting a ride, a package needing delivery). When a vehicle is assigned to a primary task, its route is not only optimized for that primary task but also simultaneously optimized for a secondary goal: the physical transport of portable data cartridges. This involves performing network theory and directed graph searches that consider both the primary objective (e.g., shortest passenger route) and the secondary objective (e.g., delivering a data cartridge to a hub along the way, or rerouting slightly to pick up a data cartridge).

Vehicles in the network are equipped to handle portable data cartridges, which are loaded and unloaded via high-bandwidth interfaces. This allows, for example, a ride-hailing driver to pick up a data cartridge from a secure data drop-off point and deliver it to a data logistics hub during their normal operational route, turning idle capacity into valuable data transport service. The method creates a symbiotic relationship between the two network types, making both more efficient and cost-effective.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive

license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating a dual-purpose logistics network, showing a vehicle performing a primary task while concurrently transporting data cartridges.

FIG. 2 is a flowchart of the dual-purpose routing optimization method.

Detailed Description of the Invention

Referring to **FIG. 1**, a dual-purpose logistics network is shown. A vehicle (101), such as a ride-hailing car, is performing its primary task, which involves moving a passenger (102) along a primary route (103). Concurrently, the vehicle is also transporting a portable data cartridge (104). The vehicle's route is optimized by a central system to achieve both the primary task (e.g., passenger pickup and drop-off) and a secondary task (e.g., transporting data cartridge 104 from an origin to a destination, potentially via a data hub (105)). This optimization leverages the existing movement of the vehicle to create an efficient data transport channel.

FIG. 2 illustrates the dual-purpose routing optimization method. The process begins with the reception of a primary transport request (201). A network optimization engine (202) identifies available vehicles (203). For each available vehicle, the engine considers its primary mission (204) and potential secondary missions, such as picking up or dropping off data cartridges (205). Using network theory and directed graph algorithms, the engine calculates an optimal route (206) that fulfills the primary request while also maximizing the achievement of secondary data logistics goals. The vehicle is then dispatched with this optimized dual-purpose route (207). As the vehicle executes its route, it performs both the primary task (208) and the data logistics task (209).

Claims

What is claimed is:

1. A method for operating a dual-purpose logistics network, the method comprising the steps of:

    - receiving a request for a primary transport task for an item or passenger between a source and a destination;
    - identifying a vehicle from a fleet of vehicles configured to perform the primary transport task and to physically transport a portable data cartridge;
    - using a computational system to determine an optimized route for the identified vehicle, said optimized route configured to fulfill the primary transport task and to concurrently facilitate a secondary data logistics task involving the physical transport of at least one portable data cartridge; and
    - dispatching the identified vehicle to execute the optimized route.

2. The method of claim 1, wherein the primary transport task is ride-hailing or package delivery.

3. The method of claim 1, wherein the secondary data logistics task comprises moving a portable data cartridge between a data pickup location and a data drop-off location.

4. The method of claim 1, wherein the portable data cartridge is loaded or unloaded via a high-bandwidth parallel interface capable of a data transfer rate exceeding 1 terabit per second.

5. The method of claim 4, wherein the high-bandwidth parallel interface is a massively parallel optical interface.

6. The method of claim 1, wherein the computational system employs network theory and directed graph searches to optimize the route for both the primary and secondary tasks.

7. A system for a dual-purpose logistics network, comprising:

    - a fleet of vehicles, each vehicle configured to perform a primary transport task and to physically transport a portable data cartridge;
    - a logistics management system configured to receive requests for primary transport tasks and to generate optimized routes for the vehicles, said optimized routes configured to fulfill the primary transport task and to concurrently facilitate a secondary data logistics task; and
    - a plurality of data pickup/drop-off locations for portable data cartridges.

8. The system of claim 7, wherein the logistics management system comprises a computational engine for performing network theory and directed graph searches.

Drawings

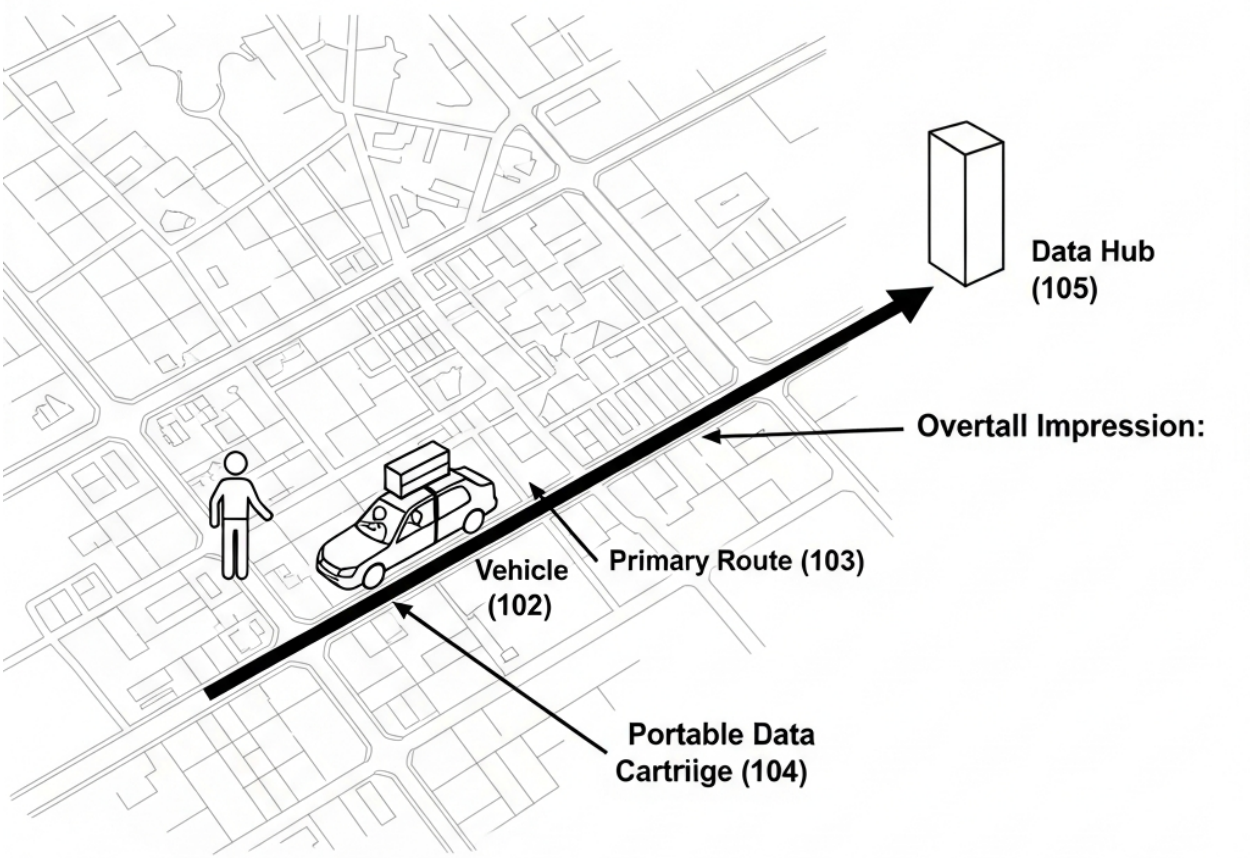


Figure 1: A conceptual diagram illustrating a dual-purpose logistics network, showing a vehicle performing a primary task while concurrently transporting data cartridges.

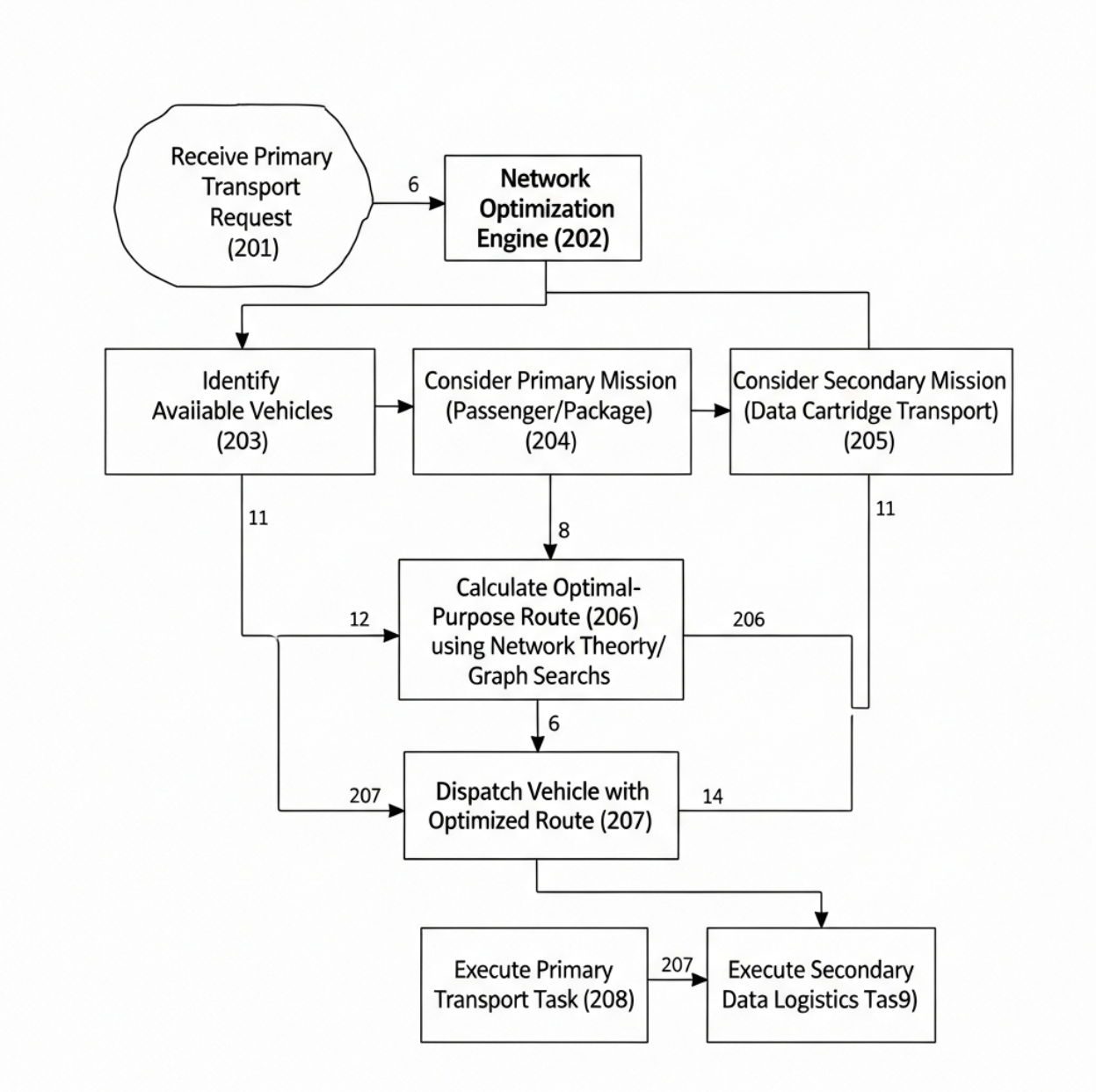


Figure 2: A flowchart of the dual-purpose routing optimization method.